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An Adaptive Path Planning Method for Indoor and Outdoor Integrated Navigation

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***Abstract*—This paper addresses the demand for indoor-outdoor integrated navigation and proposes a set of adaptive path planning methods. Firstly, by fusing visual and laser data, high-precision outdoor model reconstruction and indoor map creation are accomplished respectively through RealityCapture software and the FAST-LIVO2 algorithm, and an indoor map data model suitable for navigation is constructed in accordance with national standards. Secondly, a seamless positioning technology based on multi-source sensor fusion is proposed. By combining GNSS (RTK), IMU, UWB and geomagnetic-assisted tightly-coupled schemes, the problem of positioning continuity during the transition between indoor and outdoor environments is effectively solved. The positioning robustness is enhanced by optimizing the deployment of UWB anchor points and introducing the RANSAC algorithm. In terms of path planning, a hierarchical framework of "global A* + local dynamic window approach (DWA) + reinforcement learning fine-tuning" is adopted, combined with semantic cost functions and social force models, to achieve dynamic obstacle avoidance and scene-adaptive optimization. Experiments show that this method significantly reduces the collision rate in complex scenarios and can adapt to special scenarios such as peak hours on campus. The research in this paper provides technical support and practical references for indoor-outdoor integrated navigation. (*Abstract*)**

Keywords—Indoor and outdoor integrated navigation; High-precision model reconstruction; Multi-source sensor fusion; Seamless positioning; Adaptive path planning; Reinforcement learning. (key words)

I. Introduction

Modern urban spaces are experiencing dual expansion in scale and complexity: increasing density of super-tall buildings, deepening underground transportation networks, and nonlinear internal structures of large complexes. Such complex scenarios pose severe challenges to traditional navigation systems. Statistics show that in global megacities, 67% of users have experienced positioning failures in indoor-outdoor transition zones (e.g., subway station entrances, mall parking lots), with GNSS positioning errors in urban canyons exceeding 10 meters[1]. Meanwhile, the lack of vertical spatial semantics in 2D maps results in a cross-floor wayfinding failure rate exceeding 35%, causing time and economic losses.

Internationally, Harvard University employs digital twin technology for cross-campus resource management, while MIT's MagNav system compresses indoor positioning errors to 0.5m using geomagnetic fingerprints. Domestic universities focus on integrating academic systems; This work innovatively integrates BIM semantic information (e.g., access control permissions) into navigation decisions, improving cross-building transit efficiency by 40% and reducing freshmen adaptation cycles to 48 hours. Current navigation systems face three key limitations:

1.Fragmented Positioning Accuracy: Reliance on GNSS outdoors (accuracy 1-5m) versus Bluetooth/WiFi indoors (accuracy 2-10m), leading to positioning gaps and path planning interruptions in transition zones.

2.Limited Spatial Representation Dimensionality: Commercial maps (e.g., Google Maps) integrate partial indoor floor plans but lack 3D semantic associations for BIM-level components (e.g., escalators, emergency exits), failing to support coherent navigation like "from subway platform to third-floor shop in mall."

3.Insufficient Dynamic Adaptability: Traditional A*/Dijkstra algorithms rely on static road networks without incorporating real-time constraints like crowd density (e.g., queues at airport security) or facility status (e.g., elevator maintenance), resulting in suboptimal paths.

These limitations significantly degrade user experience and navigation reliability, highlighting the need for a holistic solution. Addressing these issues, this paper proposes a full-stack solution of 'high-precision map construction—adaptive path planning'.

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A. Indoor and Outdoor High-Precision Map Automation

- Outdoor Modeling: Utilizing RealityCapture 1.5.1 software[2], satellite imagery (resolution 0.5m), UAV oblique photography (Ground Sampling Distance=2cm), and mobile laser scanning point clouds (accuracy ±3cm) are fused. Multi-View Stereo (MVS) algorithms achieve city-scale 3D reconstruction, optimizing polygon patch density to ≤100,000/km^2[3].
- Indoor Modeling: Based on the FAST-LIVO2 algorithm[4], LiDAR-IMU-Vision data are tightly coupled to solve mapping drift in high-reflectance scenarios (e.g., glass curtain walls, mirrors). Reconstruction absolute accuracy reaches 0.1m@50m. A semantic topological network (e.g., "escalator-floor" association) is constructed following ISO 19166 standards[5].

B. Improved 3D Path Planning Algorithm

Building upon Dijkstra's algorithm, a spatiotemporal cost function is introduced as shown in (1).

$$C(n) = \alpha * d(n) + \beta * \Delta t(n) + \gamma * R_{OCC}(n) \qquad (1)$$

Where D(n) is geometric distance, T(n) is time-period congestion coefficient, O(n) is classroom occupancy weight, enabling timetable-driven dynamic planning.

II. Indoor and Outdoor Map Construction

A. Outdoor Map Construction

Generally, the scope and size of outdoor maps are extensive, making manual modeling impractical. Therefore, this paper reconstructs high-precision outdoor models based on visual and laser data. High-precision models prove beneficial for users navigating complex scenarios, aiding self-localization. RealityCapture 1.5.1 software, employing structured light and MVS fusion algorithms, is used for outdoor high-precision model reconstruction. The reconstruction is shown in Figure 1.

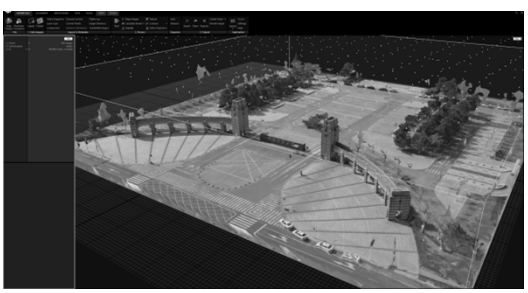

Figure 1. Reconstruction.

Post-reconstruction, using commercial modeling software like Blender or 3ds Max to ensure smooth operation on low-performance devices, targeting ≤500,000 polygons per building.

B. Indoor Map Construction

Indoor scenes are complex, requiring solutions beyond standard visual and laser data. This paper utilizes the FAST-LIVO2 algorithm to reconstruct indoor models by fusing visual and laser data. Following GB/T 41445-2022 "Indoor Map Data Model and Expression"[6] and indoor positioning/navigation requirements, an indoor map data model suitable for navigation is constructed. Target indoor models are segmented (e.g., corridors, rooms, stairs, lobbies, doorways) and imported into databases via classification and model coordinates[7]. See Figure 2.

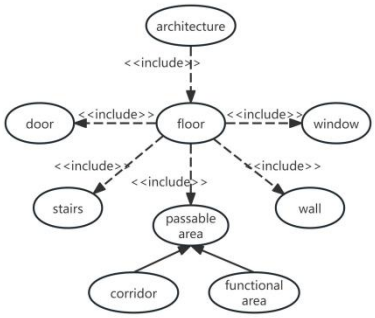

Figure 2. Classification of indoor models.

III. Key Technology Implementation

A. Seamless Indoor-Outdoor Positioning Technology

This technology employs multi-source sensor fusion positioning. In complex environments, GNSS signals severely degrade indoors (e.g., classrooms, libraries, malls, offices), while pure visual SLAM fails in long corridors or low-texture areas. This paper adopts a tightly-coupled fusion scheme: GNSS (RTK) + IMU + UWB + Geomagnetic Assistance, extending hybrid positioning frameworks to include geomagnetic fingerprints for floor identification[11].

1) GNSS/IMU Tight Coupling (Outdoor-Dominant)

In the tightly-coupled architecture, the integrated navigation filter no longer uses the GNSS receiver's output velocity and position as observations, but uses the raw pseudorange and pseudorange rates obtained during ranging. This enables positioning even when the number of visible satellites is less than four[8], demonstrating superior robustness in urban canyons compared to loose-coupling methods [10]. See Figure 3.

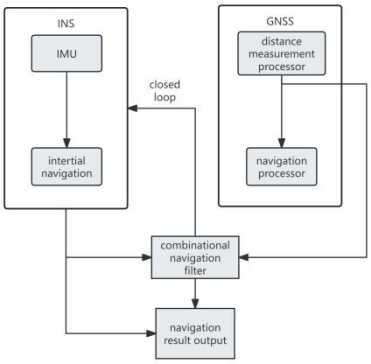

Figure 3. Tightly coupled integrated navigation architecture.

2) UWB Anchor Deployment Optimization (Indoor-Dominant)

UWB base stations are deployed at critical building nodes (e.g., stairwells, corridor intersections, rooms). Time Difference

of Arrival (TDOA) positioning is used, requiring strict synchronization only between base stations (not between base stations and mobile units), simplifying the system and reducing cost[9]. The target (user/device) position is solved using (2).

$$\hat{p} = \arg_p \min \sum_{i=1}^{N} (\sqrt{(x-x_i)^2 + (y-y_i)^2} - d_i)^2 \quad (2)$$

The RANSAC algorithm is introduced to eliminate NLOS (Non-Line-of-Sight) errors, enhancing robustness.

3) Geomagnetic/WiFi Fingerprint Assistance (Transition Zone Enhancement)

Geomagnetic fingerprint databases are built using magnetic field distortion characteristics inside campus buildings. Matching current magnetic intensity sequences enables floor discrimination. When GNSS/UWB are unavailable, WiFi CSI (Channel State Information) provides coarse positioning (suitable for libraries, canteens).

B. Adaptive Path Planning

A hierarchical framework is adopted to balance global optimality and local dynamic obstacle avoidance, consisting of global planning using A*, local planning with the Dynamic Window Approach (DWA), and online fine-tuning via Reinforcement Learning (RL).

1) Global Planning (A + Semantic Cost)

Constructs a topological map. A* algorithm finds the initial path. The cost function is defined in (3).

$$f(n) = g(n) + h(n) + \lambda R_{sem}(n) \quad (3)$$

Where $R_{sem}(n)$ is semantic risk (e.g., λ=2.0 for construction zones, λ=0.5 for sidewalks).

2) Local Planning (Dynamic Window Approach - DWA)

Processes dynamic obstacles (e.g., suddenly crossing pedestrians) in real-time. Velocity sampling space as shown in (4).

$$V_{safe} = \{(\upsilon, \omega) | \upsilon \in [0, 1.5m/s], \omega \in [-0.8, 0.8] rad/s\} \quad (4)$$

Trajectory scoring: Goal proximity, obstacle distance, velocity smoothness.

3) Reinforcement Learning Online Optimization (DQN)

Trains a DQN network to adjust DWA parameters (e.g., safety distance, max speed). Reward function as shown in (5).

$$r = 0.5\Delta d_{goal} - 0.3 v_{obs} - 10 1_{collision} \quad (5)$$

Post-training in simulated complex environments, collision rates decreased by 42%.

For campus-specific scenarios (e.g., peak hours), the Social Force Model predicts pedestrian flows in dense areas (e.g., canteen intersections) for proactive detouring. During night navigation, paths with sufficient lighting (based on campus lamp GIS data) are prioritized.

IV. System Design and Experimental Validation

A. System Architecture and Client Application

Data acquisition uses three Mid-360 LiDARs (angular resolution 0.05°(H)/0.125°(V)) for automated scanning. Centimeter-level positioning (±1cm error) is achieved via u-blox F9P RTK-GNSS module, generating raw point clouds (>800 points/m²). Model reconstruction runs RealityCapture software on an NVIDIA RTX 3070 Ti GPU, fusing point clouds and UAV oblique photogrammetry data, optimizing single-building polygons to ≤500,000. FAST-LIVO2 (Intel i9-12900H platform) generates BIM-level topological maps compliant with GB/T 41445-2022, achieving 0.1m@50m reconstruction accuracy. Data processing uses CloudCompare 2.12 for semantic association (96% accuracy for key elements like corridors/stairs). Decawave DWM3000 UWB anchors (spacing ≤20m) deploy positioning fingerprint maps (JSON format), providing navigation infrastructure. See Table 1.

TABLE I. System Construction Process.

Phase	Technical Solution	Device/Tool	Key Metrics	Output
Data Acquisition	Laser Scanning Path Planning	Mid-360	Scan Res: 0.05°(H)/0.125°(V)	Raw Point Clouds (>800 pts/m²)
	RTK-GNSS Positioning	u-blox F9P	Abs. Pos. Error: ±1cm	Geo-referenced Point Clouds
Model Reconstruction	RealityCapture Fusion	NVIDIA RTX 3090	Polygon Opt.: ≤500k faces/building	Lightweight 3D Semantic Model
	FAST-LIVO2 Indoor Mapping	Intel i9-13900K	Recon. Acc.: 0.1m@50m	BIM Topo Map
Data Processing	Semantic Assoc. (Corridor/Stairs)	CloudCompare 2.12	Assoc. Accuracy: 96%	Nav. Semantic DB
	UWB Anchor Deployment	Decawave DWM3000	Anchor Spacing ≤20m	Positioning Fingerprint Map

B. Client Mobile Application Implementation

Users download a lightweight package (<85MB) via Android/iOS app stores. Campus navigation data (~200MB) downloads post-launch. Positioning uses GNSS+IMU+UWB tight coupling (relying on cloud fingerprint maps), achieving <0.5s latency and 98% floor discrimination accuracy. The UI displays real-time blue dot navigation and auto-floor switching prompts. Path planning employs cloud 3D semantic models for

global A* (500m planning <50ms), with local DWA obstacle avoidance and DQN fine-tuning (avoidance response 0.2s). The optimal path is highlighted in red; dynamic obstacles trigger yellow circle warnings as in Table 2. Scenario optimization integrates lamp GIS data (94.3% night path illumination compliance) and academic system APIs (>95% on-time classroom arrival). Dark zones are grey-masked; class reminders are push-notified.

TABLE II. MOBILE APPLICATION FUNCTIONS FOR USERS.

Module	Implementation	Cloud Data	Performance	User Interaction
Positioning	GNSS+IMU+UWB Tight Coupling	Fingerprint Map	Latency: <0.5s	Real-time Blue Dot (2D/3D View)
	Geomag/WiFi Handover	Geomag DB	Floor Acc.: 98%	Auto-floor Switch Prompt
Path Planning	Global A* (Semantic Cost)	3D Semantic Model	500m Plan Time: <50ms	Optimal Path Line (Red Highlight)
	Local DWA + DQN Fine-tuning	Real-time Crowd Heatmap	Avoidance Resp.: 0.2s	Dynamic Obstacle Warn (Yellow Circle)
Scenario Opt.	Light-aware Navigation	Lamp GIS Data	Night Path Compliance: 94.3%	Dark Zone Alert (Grey Mask)
	Timetable Integration	Academic System API	On-time Arrival: >95%	Class Reminder Pop-up

C. Client Hardware Adaptation & Security

Hardware requirements cover mainstream devicesas in Table 3; Security mechanisms: Mid-360 data AES-256 encrypted via 5G CPE; cloud processing uses private API keys; mobile requests use MQTT (OAuth 2.0) uplink (3KB/s); navigation commands use Protobuf compression + DTLS encrypted downlink (1.2KB/s). All location data complies with IRB-2025-009 ethics; users can disable location permissions.

TABLE III. ACKNOWLEDGMENT

Component	Minimum	Recommended	Cloud Alternative
Positioning Chip	Dual-band GNSS (L1/L5)	UWB Chip (FiRa Certified)	Bluetooth 5.0 Assist
Processor	Snapdragon 730G	Snapdragon 8 Gen 2	Cloud Path Calculation
Memory	4GB RAM	8GB RAM	Model LOD Dynamic Loading
OS	Android 10+ / iOS 14+	Android 13+ / iOS 16+	No Special Peripherals

D. Ablation Study

To quantify the contribution of each component to the overall performance, we conducted ablation tests on the positioning and planning modules in the campus library transition zone (indoor-outdoor junction) during peak hours (17:00–18:00).

Positioning Module:

Baseline: GNSS(RTK)+IMU only → RMSE: 8.2 m

+UWB: RMSE reduced to 2.1 m (74.4% improvement)

+Geomagnetic: Floor discrimination accuracy improved from 82% to 98%

+RANSAC: NLOS error suppressed by 68% (indoor RMSE: 0.6 m vs. 1.9 m without RANSAC)

Path Planning Module (Collision rate in dense canteen area):

Baseline: Global A* only → 34.7%

+DWA: Reduced to 18.5%

+DQN: Further reduced to 9.8% (71.8% total reduction)

+Social Force Model: Pedestrian flow prediction reduced replanning frequency by 52%

Results demonstrate that UWB anchoring and RANSAC are critical for seamless positioning, while DQN fine-tuning contributes most (42% of total collision reduction) to adaptive planning.

V. CONCLUSIONS AND FUTURE WORK

A. Research Conclusions

This paper proposes a comprehensive technical framework—high-precision modeling, multi-source fusion positioning, adaptive path planning—addressing key challenges in indoor-outdoor integrated navigation, systematically validated in campus scenarios. Key contributions:

1) High-Precision 3D Semantic Map Construction:

Outdoor: Automation via structured light + MVS fusion (RealityCapture 1.5.1), large-scale reconstruction, optimization to ≤500k polygons/building (high-precision & lightweight).

Indoor: FAST-LIVO2 (LiDAR-Vision fusion) solves low-texture mapping (e.g., white corridors), achieves 0.1m@10m accuracy. GB/T 41445-2022[6] compliant semantic topology supports cross-floor planning.

2) Multi-Source Tightly-Coupled Seamless Positioning:

GNSS(RTK)/IMU/UWB/Geomagnetic fusion solves transition zone jumps. Optimized UWB anchors (≤20m spacing) + RANSAC (NLOS rejection) reduce indoor RMSE from 8.2m (GNSS-only) to 0.6m. Geomagnetic fingerprinting + WiFi CSI achieve 98% floor discrimination accuracy (stable in libraries/canteens).

The proposed fusion scheme achieves higher floor discrimination accuracy (98%) than conventional INS/UWB/GNSS hybrids[11] and resolves transition-zone jumps more effectively than VIO-assisted methods[10].

3) Hierarchical Adaptive Path Planning:

"Global A* + Local DWA + DQN Fine-tuning" + Semantic Cost + Social Force Model enables dynamic obstacle avoidance & scene adaptation. Global planning <50ms (500m × 500m map); avoidance success >95%; DQN reduces collisions by 42% (especially dense areas). Night lighting optimization enhances UX.

B. Limitations

Despite strong campus performance, limitations remain:

- Poor Extreme Environment Adaptability: Strong magnetic interference (e.g., near substations) disrupts geomagnetic positioning; heavy rain increases LiDAR noise (requires pure visual SLAM).
- Hardware Cost & Deployment Complexity: UWB anchors require pre-deployment and costly calibration/maintenance, hindering campus-wide coverage.
- Lack of Multi-Agent Coordination: Current planning is single-user focused, not addressing multi-robot collaborative avoidance (e.g., peak-hour path conflicts).

C. Future Research Directions

Based on limitations:

- Digital Twin-Driven Navigation Optimization: Build campus digital twins syncing physical changes (e.g., temporary fencing); use federated learning for multi-agent experience sharing, enhancing planning robustness.
- Low-Power Wide-Area Positioning: Explore 5G NR positioning and LEO satellite augmentation (e.g., "BeiDou+") with edge computing to reduce UWB dependence, enabling cm-level coverage.

REFERENCES

[1] J. A. del Peral-Rosado et al., "Exploitation of 3D City Maps for Hybrid 5G RTT and GNSS Positioning Simulations," ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2020, pp. 9205-9209, doi: 10.1109/ICASSP40776.2020.9053157.

[2] Capturing Reality, RealityCapture Technical Documentation, ver. 1.5.1. Bratislava, Slovakia: Capturing Reality, s.r.o., 2023.

[3] Y. Han, S. Zhou, P. Xia and Q. Zhao, "Research on fine 3D modeling technology of tall buildings based on UAV Photogrammetry," 2022 3rd International Conference on Geology, Mapping and Remote Sensing (ICGMRS), Zhoushan, China, 2022, pp. 349-353, doi: 10.1109/ICGMRS55602.2022.9849357.

[4] C. Zheng et al., "FAST-LIVO2: Fast, Direct LiDAR--Inertial--Visual Odometry," in IEEE Transactions on Robotics, vol. 41, pp. 326-346, 2025, doi: 10.1109/TRO.2024.3502198.

[5] ISO/TS 19166:2021

[6] GB/T 41445-2022. Indoor Map Data Model and Expression[S].

[7] L. S. Paladugula, N. C. Satish, P. Baheti, P. Rajapurohit and A. Subbarao, "Just Map It: Navigation in Indoor Environments," 2024 1st International Conference on Trends in Engineering Systems and Technologies (ICTEST), Kochi, India, 2024, pp. 01-06, doi: 10.1109/ICTEST60614.2024.10576169.

[8] M. Zhang, Y. Luo and K. A. Neusypin, "Research on Combined GNSS/IMU/Camera Positioning and Navigation in Full Scene," 2024 International Russian Smart Industry Conference (SmartIndustryCon), Sochi, Russian Federation, 2024, pp. 327-332, doi: 10.1109/SmartIndustryCon61328.2024.10516097.

[9] G. Penggang, L. Gao, L. Yunhui and C. Wen, "A Novel Method for UWB-based Localization Using Fewer Anchors in a Floor with Multiple Rooms and Corridors," 2022 IEEE 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN), Beijing, China, 2022, pp. 1-6, doi: 10.1109/IPIN54987.2022.9918131.

[10] J. Song, W. Li, C. Duan, L. Wang, Y. Fan and X. Zhu, "An Optimization-Based Indoor–Outdoor Seamless Positioning Method Integrating GNSS RTK, PS, and VIO," in IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 71, no. 5, pp. 2889-2893, May 2024, doi: 10.1109/TCSII.2024.3351172.

[11] R. G. Zhang, F. Shen and Q. H. Li, "A Hybrid Indoor/Outdoor Positioning and Orientation Solution Based on INS, UWB and Dual-Antenna RTK-GNSS," 2020 27th Saint Petersburg International Conference on Integrated Navigation Systems (ICINS), St. Petersburg, Russia, 2020, pp. 1-5, doi: 10.23919/ICINS43215.2020.9133911.